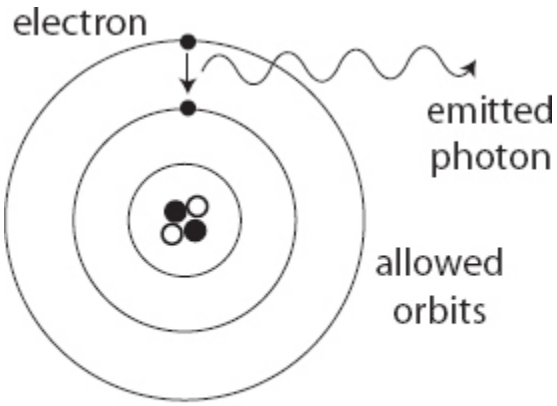
hundred-billionth of a second. If ordinary matter made of atoms only lasted for that long, someone would have noticed by now.

This bothered a lot of people, most notably Niels Bohr, who had briefly worked under Rutherford in 1912. In 1913, Bohr published a series of three papers, later known simply as “the trilogy,” in which he put forth another of those audacious, out-of-the-blue ideas that characterized the early years of quantum theory. What if, he asked, electrons can’t spiral down into atomic nuclei because electrons simply aren’t allowed to be in any orbit they want, but instead have to stick to certain very specific orbits? There would be a minimum-energy orbit, another one with somewhat higher energy, and so on. But electrons weren’t allowed to go any closer to the nucleus than the lowest orbit, and they weren’t allowed to be in between the orbits. The allowed orbits were quantized.



Electrons can jump between allowed orbits in the Bohr atom

Bohr’s proposal wasn’t quite as outlandish as it might seem at first. Physicists had studied how light interacted with different elements in their gaseous form—hydrogen, nitrogen, oxygen, and so forth. They found that if you shined light through a cold gas, some of it would be absorbed; likewise, if you passed electrical current through a tube of gas, the gas would start glowing (the principle behind fluorescent lights still used today). But they only emitted and absorbed certain very specific frequencies of light, letting other colors pass right through. Hydrogen, the simplest element with just a single proton and a single electron, in particular had a very regular pattern of emission and absorption frequencies.

For a classical Rutherford atom, that would make no sense at all. But in Bohr’s model, where only certain electron orbits were allowed, there was an